Olga-Dimitra Stoura
olgastoura@gmail.com | 413-472-6889 (US number) | +306948712233 (Greek number)

EDUCATION

University of Massachusetts Amherst September 2023-May 2027 Amherst, MA
Bachelor of Science in Mechanical Engineering

- Relevant Coursework: Heat transfer, System dynamics, Design of mechanical components (design and manufacture a multifunctional wrench head), Thermodynamics, Statics, Fluids, Dynamics, Material Science, Fndm of Electrical Engineering (Car demo project), Mechatronics, Manufacturing processes.

EXPERIENCE

Shrimathveeravalli Research Group October 2025-Present Umass Amherst, Massachusetts
Laboratory intern

- Fabrication of blood vessels, 2D flow and screening micro chips.
- Research in pancreatic endoscopy-CAD design of biopsy needle.

Stouras EE May 2025-August 2025 Komotini, Greece
Consulting Intern

- Collaborated on site with a team of five to consult with electrical and mechanical. engineers during construction of a hospital
- Conducted safety checks on materials to ensure proper use up to code.
- Utilized Matlab to check pipe pressure and electrical current through power sources.
- Visualized the renderings of the hospital using Autocad and solidworks.

Alexandroupoli Beaches Summer 2024, 2025 Alexandroupoli, Greece
Lifeguard

- Executed daily set up, including putting up flags and buoys, filling the ATV, checking the oxygen tanks in case of an emergency and prepping the first aid kit to ensure preparedness for all lifeguards during the day.
- Utilized First Aid and CPR certification in high pressure situations.
- Monitored the beach during shifts to ensure that individuals were safe and compliant with the beach rules .
- Communicated with the team through radio to connect on any issues over large distances.

PROJECTS

Autonomous Vehicle Design and Demonstration
ECE 361 | University of Massachusetts Amherst | [Fall 2025]

- Independently designed, programmed, and assembled an autonomous vehicle using Arduino, sensors, and motors.
- Developed and implemented a sensor-based control system to detect obstacles and execute 90-degree turns in response.
- Integrated hardware and software components, troubleshooting system performance to achieve autonomous navigation.

Multifunctional Wrench Design and Development

MIE 313 – Design of Mechanical Components | University of Massachusetts Amherst | Spring 2026

- Collaborated with a team to design a lightweight, multifunctional wrench handle featuring a detachable crowfoot head, screwdriver-bit storage, and bottle opener.
- Developed and refined SolidWorks CAD models through iterative design, applying mechanical design principles and factor of safety calculations.
- Fabricated and tested ABS prototypes using FDM 3D printing, evaluating structural performance and identifying design improvements.

ATHLETICS

University of Massachusetts Amherst September 2023-Present Amherst, MA

Division 1 student-athlete

- Manage a disciplined schedule of 20+ hours a week of training, lifting, rehabilitation, and nutrition on top of a full academic course load.
- Work with a team in achieving a common goal by supporting one another.